Who Speaks Matters More Than How: Automated Accounts, Account Reach, and Message Features in COVID-19 Vaccine Tweets Around the Delta Wave

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Abstract

Studies of crisis communication on social media often infer which message features drive engagement from large public tweet corpora, yet such corpora can contain automated accounts and are rarely adjusted for who is posting. We re-examine Media Richness Theory and the feature-based operationalization of Dialogic Communication Theory using 180,278 deduplicated English-language COVID-19 vaccine tweets from a public Kaggle corpus, covering windows before (April–June 2021), during (July–September 2021) and after (October–November 2021) the Delta wave. Automated vaccine-slot notification accounts produced 34.0% of Delta-window tweets and attracted almost no engagement. A naive negative binomial model of all Delta tweets suggested that a link or attached media multiplied engagement 4.71-fold. After removing automated accounts, adjusting for account reach and clustering standard errors by account, the association fell to 1.40 (95% CI 1.19–1.64). Positive sentiment remained positively associated with engagement (IRR = 1.48), questions (0.85) and hashtags (0.91 per tag) were modestly negative, and @mentions were unrelated to engagement. Follower count was the strongest predictor: doubling an account's followers was associated with about 36% more engagement. Across windows, the link and sentiment associations were stable in direction, whereas the dialogic cues shifted, and we found no evidence that the link effect reverses after the peak. The results suggest that who speaks matters more than how a message is designed, that counting dialogic features captures little of what dialogue theory describes, and that auditing corpora for automation is a precondition for drawing communication lessons from public tweet data.

Keywords: COVID-19 vaccines; Twitter; automated accounts; media richness theory; dialogic communication; engagement; negative binomial regression; computational social science

1. Introduction

During the COVID-19 pandemic, public health agencies, journalists and ordinary users discussed vaccines on the same platforms, and Twitter became one of the most closely studied arenas of that conversation (Cinelli et al., 2020; Lyu et al., 2021; Xu et al., 2022; Yousefinaghani et al., 2021). For

communicators, the practical question is which message choices earn attention. Two theoretical traditions give competing answers. Media Richness Theory (MRT) holds that under uncertainty, messages carrying more cues, such as a link to a source or an image, should be favoured (Daft & Lengel, 1986). Dialogic Communication Theory (DCT), as operationalized for Twitter, treats questions, hashtags and @mentions as invitations to two-way exchange that should increase engagement (Kent & Taylor, 1998; Rybalko & Seltzer, 2010).

Evidence from official government accounts during the pandemic is mixed. For the Healthy China account on Sina Weibo, posts with pictures or video drew less engagement than plain text, while dialogic features drew more (Chen et al., 2020). For government accounts in Hubei Province, both media richness and dialogic loops were modestly positive, and the effects differed across stages of the outbreak (Zhang et al., 2022). Much less is known about the broader, user-generated vaccine conversation, where most posts do not come from institutions.

Studying that broader conversation usually means analysing large public tweet corpora, which raises two problems that the engagement literature often leaves aside. The first is automation. Automated accounts can make up a large share of a topical corpus, post in templated formats and receive little or no engagement (Broniatowski et al., 2018). If their templates happen to include links or hashtags, their presence alone can manufacture associations between message features and engagement. The second problem is reach. How much engagement a tweet receives depends heavily on who posts it (Suh et al., 2010), and the accounts with large audiences may write differently from everyone else. A model without account-level controls therefore confounds the message with the messenger. A related statistical issue is that many tweets come from the same accounts, so standard errors that treat tweets as independent are too small (Cameron & Miller, 2015).

This study addresses both problems using a widely available Kaggle corpus of English-language COVID-19 vaccine tweets (Preda, 2021), focusing on the Delta wave of July–September 2021 and on the periods immediately before and after it. The Delta period combined vaccine availability with breakthrough infections, booster debates and continuing safety discussion, which makes it a useful setting for testing whether message-feature effects hold beyond the first months of the vaccine rollout. We ask three research questions. RQ1: How prevalent are automated accounts in the corpus, and how do automation and account reach change the estimated associations between message features and engagement? RQ2: Once automated accounts are removed and account reach is accounted for, which message features are associated with engagement during the Delta wave? RQ3: Are these associations stable across the windows before, during and after the Delta wave?

The study makes three contributions. Methodologically, it shows that a third of the Delta-window corpus came from automated vaccine-slot accounts and that ignoring them, together with ignoring account reach, inflates the apparent link effect more than threefold and distorts the dialogic estimates. Empirically, it provides account-adjusted estimates of message-feature associations for user-generated vaccine discourse across three windows, using a single pooled interaction model to test change between windows. Theoretically, it qualifies both frameworks: the advantage of informational cues is real but modest, and feature-based dialogic cues do not behave as DCT would predict, which is consistent with critiques that such counts do not measure dialogue (Taylor & Kent, 2014; Wirtz & Zimbres, 2018).

2. Theoretical Background

2.1 Media richness and informational cues

MRT was developed to explain how organizations match media to tasks: richer media, which carry more cues, allow fast feedback and support personal focus, are better suited to equivocal situations (Daft & Lengel, 1986). On a text-first platform such as Twitter, the most common way to add cues to a post is to attach a photo or video or to link to an external article. Applied to crisis communication, the MRT expectation is that audiences facing uncertainty reward posts that offer more information than the text alone. The evidence for official accounts is not uniform. Chen et al. (2020) found that pictures and video reduced engagement with the Healthy China account, whereas Zhang et al. (2022) found small positive media-richness effects for Hubei government accounts. For general vaccine discourse, we test the MRT expectation directly.

H1: Tweets that contain a link or attached media receive more engagement than tweets that do not.

2.2 Dialogic communication and its feature-based operationalization

Kent and Taylor (1998) proposed principles for building dialogic relationships between organizations and publics online, including a dialogic loop through which publics can ask questions and receive responses. Rybalko and Seltzer (2010) translated this approach to Twitter, treating questions, hashtags and @mentions as markers of a dialogic orientation. Much subsequent research has followed this feature-based route, and the implied prediction is that tweets carrying these cues should attract more engagement.

This operationalization has been criticized from within the dialogic tradition. Dialogue, in Kent and Taylor's account, is a relational orientation and a form of engagement between parties, not the presence of particular interface or message features (Taylor & Kent, 2014). Reviews of two decades of digital dialogic research have noted that studies mostly measure whether such features are present and have rarely linked them to dialogic outcomes (Sommerfeldt & Yang, 2018; Wirtz & Zimbres, 2018). DCT was also formulated for organization–public relationships, whereas the corpus studied here is dominated by

individual users. We therefore test the narrower, widely used proposition that dialogic cues increase engagement, and we interpret the result as evidence about that operationalization rather than about dialogue itself.

H2: Questions, hashtags and @mentions are positively associated with engagement.

2.3 Emotional tone

Evidence on emotional tone is mixed. For Hubei government accounts, more positive emotional valence was associated with lower engagement in two of three outbreak stages (Zhang et al., 2022), and for the Healthy China account, valence moderated the effects of media richness and dialogic features (Chen et al., 2020). Because these findings come from official accounts and the direction for user-generated vaccine tweets is not established, we treat sentiment as a research question within RQ2 rather than stating a directional hypothesis.

2.4 Automation and account reach as confounders

Automated and semi-automated accounts are a known feature of vaccine discussion on Twitter (Broniatowski et al., 2018). From a measurement perspective, they matter for engagement research for a simple reason: they tend to post large volumes of templated content that few people interact with. If a corpus contains many such tweets and their templates differ systematically in links, hashtags or questions, any regression of engagement on these features partly measures the difference between automated and human posting. Account reach poses a parallel problem. Large-scale analysis of retweeting has shown that the number of followers and followees is strongly associated with how widely a tweet spreads (Suh et al., 2010). Because the kinds of accounts that have large audiences, such as news organizations and public figures, also tend to use particular message styles, omitting reach attributes the messenger's audience to the message's design.

2.5 Crisis phases

Crisis communication models describe crises as unfolding in stages with different communication needs (Reynolds & Seeger, 2005), and empirical work on official accounts has found that engagement drivers differ across outbreak stages (Zhang et al., 2022). The Delta wave offers a natural sequence of a pre-wave period, a peak period and a post-wave period, which allows us to ask whether message-feature associations change with the phase of the crisis (RQ3).

3. Method

3.1 Data

We used the All COVID-19 Vaccines Tweets corpus published on Kaggle under a CC0 public-domain licence (Preda, 2021). The file contains 228,207 English-language tweets posted between 12 December 2020 and 23 November 2021 that mention COVID-19 vaccines, together with tweet text, timestamp, like and retweet counts, the posting application (source) and account metadata, including follower and friend counts, verification status and account creation date. The corpus contains no retweets. The collector's query parameters are not documented beyond the vaccine names, so the corpus should be treated as a convenience sample of vaccine discourse.

We defined three windows around the Delta wave: pre-Delta (1 April–29 June 2021), Delta (1 July–29 September 2021) and post-Delta (1 October–23 November 2021, the end of the corpus). Within each window, exact duplicate texts and tweets with missing text were removed, leaving 77,745, 71,845 and 30,688 tweets, respectively (180,278 in total). The corpus has collection gaps: no tweets were recorded between 23 April and 12 May 2021 or between 25 and 27 October 2021, and fewer than 20 tweets per day were recorded between 2 and 6 July 2021 (Figure 1).

3.2 Identifying automated accounts

An exploratory topic model of the Delta window, estimated with BERTopic (Grootendorst, 2022) on 65,290 cleaned unique texts, produced three large clusters of templated vaccination-booking notifications containing 24,966 texts (38.2%), with top terms such as age, 18, pincode, dose2, availability and Bengaluru. These templates came from accounts that automatically announce vaccination slots on India's CoWIN platform. We therefore classified tweets as automated when their source label referred to CoWIN or vaccine-slot alerting (for example, CowinAlertsBot, Cowin Vaccination Availability, VaxBlr and cowin_vaccine_app) or contained the string bot, excluding the Tweetbot client used by human users.

The rule flagged 24,415 Delta-window tweets (34.0%) posted under eight display names. Five vaccine-slot accounts produced 24,404 of them (two of these display names differ only by a space and may belong to one account), and the remaining 11 came from three other bot applications. As a validity check, every one of the 24,404 vaccine-slot tweets contained booking terms (slot, pincode, 18+ or 45+, dose 1 or dose 2, or CoWIN), compared with 1.7% of the unflagged tweets. The rule flagged 8,395 pre-Delta tweets (10.8%; 14 display names) and 4,408 post-Delta tweets (14.4%; nine display names). Because it relies on source labels, the rule is conservative: automated accounts that post through standard Twitter clients are not detected.

3.3 Measures

Engagement was the sum of a tweet's likes and retweets as recorded in the corpus; replies and quote counts are not available. Link or attached media was a binary indicator of a URL token (http) in the tweet text. In the tweet text format used by the corpus, both external links and natively attached photos or videos appear as shortened t.co links, so this indicator marks either and cannot separate them. Following Rybalko and Seltzer (2010), dialogic cues were measured as the presence of at least one question mark, the number of hashtags and the number of @mentions. Sentiment was the VADER compound score, which ranges from −1 to +1 (Hutto & Gilbert, 2014). Account-level controls were log(1 + followers), log(1 + friends), verification status and account age in years at the time of posting. Tweet-level controls were tweet length in units of ten characters and a weekend indicator.

3.4 Analysis

Engagement counts were overdispersed, so we estimated negative binomial (NB2) regressions by maximum likelihood (Hilbe, 2011) in statsmodels, with standard errors clustered by account (Cameron & Miller, 2015). Because the corpus provides no user identifiers, accounts were identified by their display names. For the Delta window we estimated a sequence of four models. M0 is the naive specification: all tweets, message features and tweet length, and conventional standard errors. M1 adds the account-level controls and weekend indicator to M0, with clustered standard errors, using all tweets. M2 removes automated accounts but omits the account controls. M3 removes automated accounts and includes all controls; it is our preferred specification. Comparing the four models shows how much of the naive estimate is due to automation, to omitted reach, or to both.

We ran three robustness checks on M3: excluding the 52 accounts that posted more than 50 tweets in the window (R1), keeping one randomly selected tweet per account (R2), and replacing the NB2 model with a zero-inflated negative binomial model whose inflation component depended on followers and tweet length (R3; Lambert, 1992). To answer RQ3, we estimated M3 separately for each window and then fitted a single pooled model to all non-automated tweets from the three windows. In it, every predictor was interacted with window indicators, the Delta window served as the reference, and standard errors were clustered by account because some accounts posted in more than one window. The interaction terms test whether an association differs between windows. We report incidence rate ratios (IRRs) with 95% confidence intervals. The full code and model outputs are publicly available (see Data Availability).

4. Results

4.1 Corpus composition

Automated accounts were concentrated in the Delta window (Figure 1). Their tweets received a mean of 0.06 engagements, and 93.7% received none, compared with a mean of 9.74 for the remaining Delta tweets. Removing them changed the descriptive profile of the window substantially (Table 1). Mean engagement rose from 6.45 to 9.74, the share of tweets with no engagement fell from 61.9% to 45.5%, and the share of tweets containing a question rose from 5.1% to 7.8%. Among the 47,430 non-automated Delta tweets from 23,963 accounts, 86.2% contained a link or attached media, tweets carried 2.01 hashtags and 0.59 @mentions on average, and 8.5% came from verified accounts. The median account posted one tweet in the window and had 346 followers. Using VADER thresholds of ±0.05, 36.8% of these tweets were positive, 44.5% neutral and 18.8% negative.

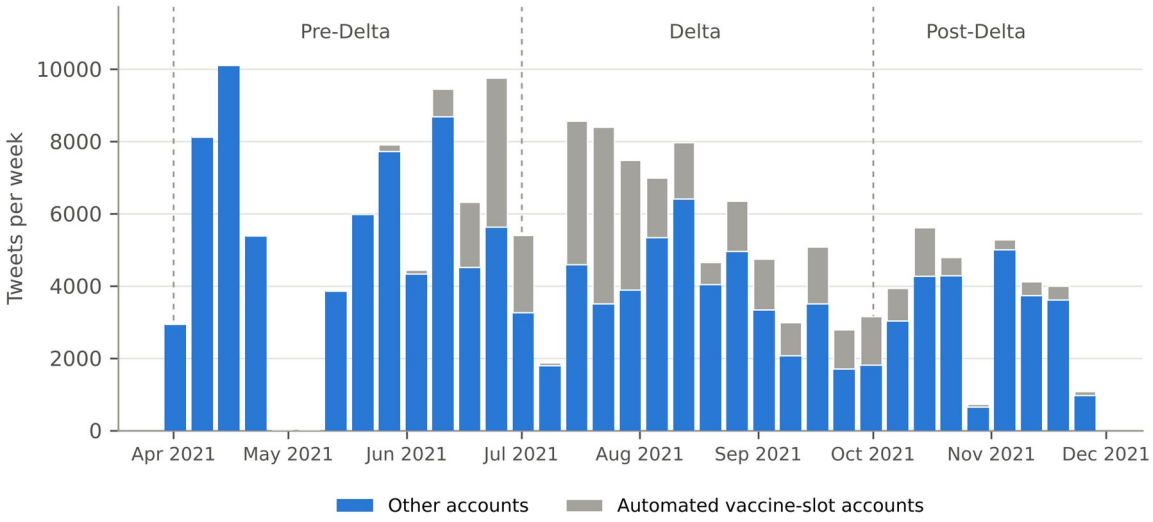


Figure 1. Weekly volume of deduplicated vaccine tweets in the three windows, split into automated vaccine-slot accounts and all other accounts. Dashed lines mark window starts. Weeks with no bars reflect gaps in the original data collection.

Table 1. Descriptive statistics by window, for all deduplicated tweets and excluding automated accounts.

	Pre-Delta: all	Pre-Delta: excl. automated	Delta: all	Delta: excl. automated	Post-Delta: all	Post-Delta: excl. automated
Automated share (%)	10.8	—	34.0	—	14.4	—
Tweets	77,745	69,350	71,845	47,430	30,688	26,280
Accounts	37,498	37,488	23,971	23,963	13,212	13,203
Mean engagement	16.06	17.97	6.45	9.74	15.78	18.38
Median engagement	1	1	0	1	1	1
No engagement (%)	44.6	40.9	61.9	45.5	43.0	37.4
Link or attached media (%)	77.5	85.5	82.5	86.2	86.5	84.2
Question (%)	7.3	8.2	5.1	7.8	7.2	8.4
Hashtags (mean)	1.85	1.84	1.77	2.01	1.93	1.89
@Mentions (mean)	0.49	0.55	0.39	0.59	0.67	0.79
VADER sentiment (mean)	0.093	0.104	0.118	0.104	0.180	0.140
Verified accounts (%)	9.4	10.6	5.6	8.5	8.6	10.0

Note. Engagement is likes plus retweets. Windows: pre-Delta 1 April–29 June 2021; Delta 1 July–29 September 2021; post-Delta 1 October–23 November 2021.

4.2 How automation and account reach change the estimates

Table 2 and Figure 2 report the Delta-window models. The naive model M0 reproduces the pattern of large message-feature effects. A link or attached media was associated with 4.71 times more engagement (95% CI 4.37–5.07), and questions (IRR = 0.69), hashtags (0.94 per tag) and @mentions (0.94 per mention) were all negatively associated with engagement (all $p < .001$). Each correction changed this picture. Adding account controls while keeping automated tweets (M1) reduced the link estimate to 2.97 and made the dialogic estimates null or positive: questions 1.15 ($p = .24$), hashtags 1.02 ($p = .78$) and @mentions 1.19 ($p < .001$). Removing automated accounts without controls (M2) reduced the link estimate to 2.03 but strengthened the negative dialogic estimates (questions 0.45, hashtags 0.83, @mentions 0.77). The direction of the dialogic estimates therefore depends on which correction is applied, which shows how strongly automation and reach confound them.

In the preferred model M3, a link or attached media was associated with 40% more engagement (IRR = 1.40, 95% CI 1.19–1.64), supporting H1, although the association is far smaller than the naive estimate. Positive sentiment was positively associated with engagement: moving from a neutral to a maximally positive VADER score corresponded to an IRR of 1.48 (95% CI 1.28–1.70). Questions were associated with 15% less engagement (IRR = 0.85, 95% CI 0.75–0.97, $p = .015$) and each hashtag with 9% less (IRR = 0.91, 95% CI 0.87–0.95), while @mentions were unrelated to engagement (IRR = 1.03, 95% CI 0.98–1.08, $p = .21$). H2 is therefore not supported. Account reach dominated the model. Each unit increase in log followers was associated with an IRR of 1.55 (95% CI 1.49–1.63), which corresponds to roughly 36% more engagement per doubling of followers. Older accounts received slightly less engagement (IRR = 0.94 per year), and the verification estimate was positive but imprecise (IRR = 1.39, 95% CI 0.94–2.04). Adding the controls to M2 improved fit by about 15,000 AIC points.

Table 2. Negative binomial models of engagement in the Delta window: incidence rate ratios [95% CI].

	M0 Naive	M1 + Account controls	M2 − Automated	M3 Corrected
Message features				
Link or attached media	4.71*** [4.37, 5.07]	2.97** [1.53, 5.74]	2.03*** [1.36, 3.04]	1.40*** [1.19, 1.64]
Sentiment (VADER)	1.37*** [1.28, 1.46]	1.65*** [1.31, 2.08]	1.48*** [1.19, 1.84]	1.48*** [1.28, 1.70]
Question	0.69*** [0.62, 0.76]	1.15 [0.91, 1.47]	0.45*** [0.35, 0.58]	0.85* [0.75, 0.97]
Hashtags (count)	0.94*** [0.93, 0.96]	1.02 [0.90, 1.15]	0.83*** [0.77, 0.90]	0.91*** [0.87, 0.95]
@Mentions (count)	0.94*** [0.92, 0.97]	1.19*** [1.08, 1.31]	0.77*** [0.73, 0.82]	1.03 [0.98, 1.08]
Controls				
Tweet length (10 chars)	0.94*** [0.93, 0.95]	0.95* [0.92, 0.99]	1.04 [1.00, 1.09]	1.01 [0.99, 1.03]
Log followers	—	1.40*** [1.31, 1.50]	—	1.55*** [1.49, 1.63]
Log friends	—	1.44*** [1.22, 1.71]	—	0.92* [0.85, 0.99]
Verified	—	2.32*** [1.48, 3.63]	—	1.39 [0.94, 2.04]
Account age (years)	—	0.96** [0.93, 0.99]	—	0.94*** [0.92, 0.96]
Weekend	—	1.06 [0.92, 1.23]	—	1.07 [0.93, 1.23]
Dispersion (α)	8.92	4.48	5.60	3.63
Automated accounts	Included	Included	Excluded	Excluded
Tweets (accounts)	71,845 (23,971)	71,845 (23,971)	47,430 (23,963)	47,430 (23,963)
Standard errors	Conventional	Clustered	Clustered	Clustered
AIC	259,082.2	232,174.2	225,624.8	210,618.4

Note. Clustered standard errors are clustered by account (display name). AIC is comparable only between models fitted to the same tweets (M0 with M1; M2 with M3). Confidence bounds are rounded to two decimals, so a bound shown as 1.00 may lie slightly below 1; significance stars follow the unrounded p values. *p < .05. **p < .01. ***p < .001.

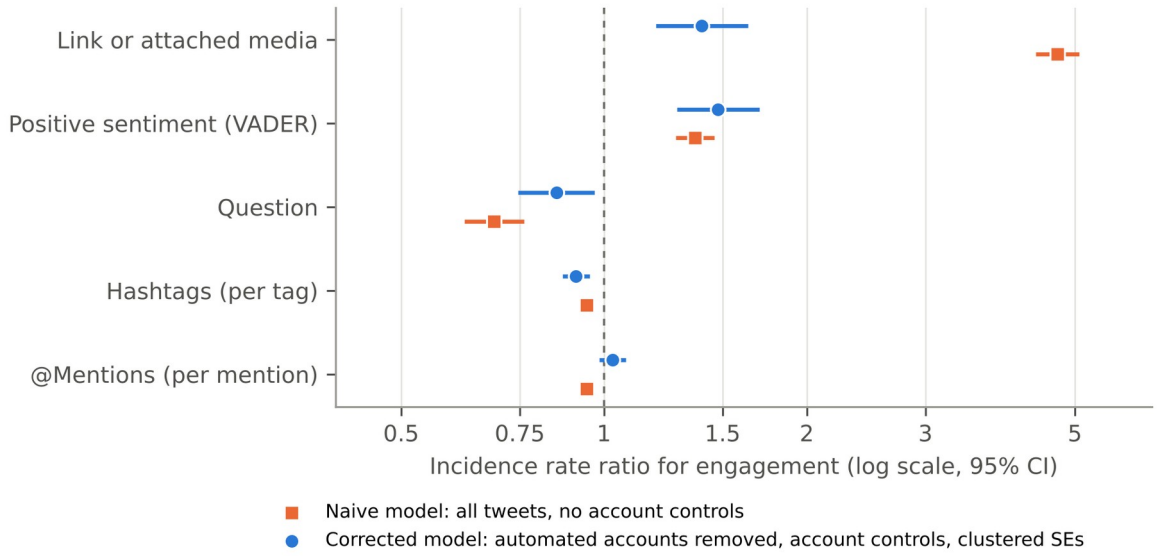


Figure 2. Incidence rate ratios for message features in the Delta window under the naive model (M0: all tweets, no account controls, conventional standard errors) and the corrected model (M3: automated accounts removed, account controls, standard errors clustered by account). Bars are 95% confidence intervals; the x-axis is logarithmic.

4.3 Robustness

The M3 estimates were stable across the robustness checks (Table 3). Excluding high-volume accounts (R1) and keeping one tweet per account (R2) left the link (IRR 1.45 and 1.62), sentiment (1.49 and 1.51), question (0.82 and 0.76) and hashtag (0.89 and 0.94) estimates in the same direction and of similar size, and @mentions remained unrelated to engagement in both. The zero-inflated model (R3) gave practically

identical count estimates and did not improve fit (AIC 210,623.2 versus 210,618.4 for NB2), and none of its inflation terms was significant. Once automated accounts were removed and reach was controlled, a standard negative binomial model was adequate.

Table 3. Robustness checks for the corrected Delta-window model (M3): incidence rate ratios [95% CI].

	R1 Accounts ≤ 50 tweets	R2 One tweet per account	R3 Zero-inflated NB
Link or attached media	1.45*** [1.23, 1.70]	1.62*** [1.50, 1.76]	1.40*** [1.19, 1.64]
Sentiment (VADER)	1.49*** [1.31, 1.70]	1.51*** [1.41, 1.62]	1.48*** [1.28, 1.70]
Question	0.82** [0.72, 0.95]	0.76*** [0.69, 0.84]	0.85* [0.74, 0.97]
Hashtags (count)	0.89*** [0.87, 0.92]	0.94*** [0.92, 0.95]	0.91*** [0.87, 0.95]
@Mentions (count)	1.01 [0.97, 1.05]	1.02 [1.00, 1.05]	1.03 [0.98, 1.08]
Tweets	41,221	23,964	47,430
AIC	182,054.6	99,161.3	210,623.2 (NB2: 210,618.4)

Note. All models exclude automated accounts and include the account and tweet-level controls of M3. R1 and R3 use standard errors clustered by account; R2 keeps one randomly selected tweet per display name (plus one tweet with no display name) and uses conventional standard errors. Confidence bounds are rounded to two decimals; stars follow the unrounded p values. *p < .05. **p < .01. ***p < .001.

4.4 Change across windows

Figure 3 and Table 4 compare the corrected associations across windows. The link association was positive before (IRR = 1.60) and during (1.40) the Delta wave and smaller and imprecise afterwards (1.14, 95% CI 0.85–1.55). None of the differences from the Delta window was significant (pre-Delta p = .16; post-Delta p = .24). Positive sentiment was positively associated with engagement in all three windows (IRRs 1.56, 1.48 and 1.24), again without significant differences. The dialogic cues were less stable. Questions carried their largest penalty before the wave (IRR = 0.66), significantly larger than during it (p = .006), and were unrelated to engagement afterwards. Hashtags were unrelated to engagement before the wave and negative during and after it; the pre-Delta difference was marginal (p = .056). @Mentions moved from a negative association before the wave (IRR = 0.91; difference from Delta p < .001) to no association during it and a small positive association after it (IRR = 1.07), although the Delta–post-Delta difference was not significant (p = .15). In short, the data show no reversal of the link effect after the peak; what varies across phases is the role of the dialogic cues.

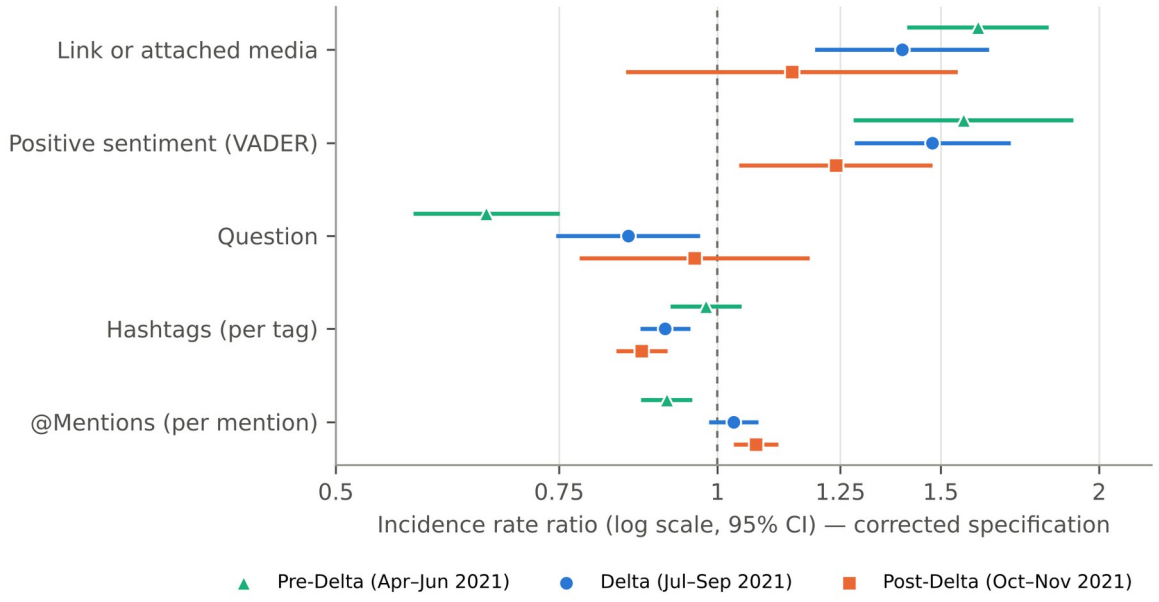


Figure 3. Incidence rate ratios for message features in the corrected specification, estimated separately for each window. Bars are 95% confidence intervals clustered by account; the x-axis is logarithmic.

Table 4. Corrected models by window: incidence rate ratios [95% CI] and tests of difference from the Delta window.

	Pre-Delta	Delta	Post-Delta	p: Pre vs. Delta	p: Post vs. Delta
Link or attached media	1.60*** [1.41, 1.82]	1.40*** [1.19, 1.64]	1.14 [0.85, 1.55]	.164	.244
Sentiment (VADER)	1.56*** [1.28, 1.91]	1.48*** [1.28, 1.70]	1.24* [1.04, 1.48]	.648	.112
Question	0.66*** [0.58, 0.75]	0.85* [0.75, 0.97]	0.96 [0.78, 1.18]	.006	.330
Hashtags (count)	0.98 [0.92, 1.04]	0.91*** [0.87, 0.95]	0.87*** [0.83, 0.91]	.056	.111
@Mentions (count)	0.91*** [0.87, 0.95]	1.03 [0.98, 1.08]	1.07*** [1.03, 1.12]	< .001	.154
Tweets (excl. automated)	69,350	47,430	26,280		

Note. Each window model uses the M3 specification. p values for differences come from a pooled model of all non-automated tweets with every predictor interacted with window indicators (Delta = reference; N = 143,060), with standard errors clustered by account. *p < .05. **p < .01. ***p < .001.

5. Discussion

5.1 Automation as a measurement problem

The most consequential finding is methodological. A third of the Delta-window corpus consisted of templated booking notifications from a handful of automated accounts, and these tweets were almost never engaged with. Included in a naive model, they made a link or attached media appear to multiply engagement almost fivefold and produced dialogic estimates whose sign depended on which other corrections were made. None of this reflects how people respond to vaccine messages. The problem is not specific to this corpus. Any keyword-based collection about a public service, such as appointments, alerts or service status, can attract automated announcement accounts. We therefore recommend that engagement studies built on public tweet corpora report how they screened for automation, inspect source

labels and templated clusters (topic models were useful here as a diagnostic), and show how estimates change when automated content is removed.

5.2 Who speaks matters more than how

Among non-automated tweets, account reach was by far the strongest predictor of engagement, consistent with earlier large-scale evidence on retweeting (Suh et al., 2010). Controlling for reach did not simply add precision; it changed the substantive conclusions, most visibly for @mentions. For practitioners, this means that message design operates within the audience an account has already built. A link or a positive framing can raise expected engagement by a few tens of percent, whereas under our model an account with a thousand times more followers can expect roughly twenty times more engagement. For researchers, it means that message-feature effects estimated without account controls should be treated as upper bounds that partly describe who is speaking.

5.3 Media richness: a modest and stable advantage

H1 was supported, but in a far weaker form than the naive estimate suggests. Tweets with a link or attached media received about 40% more engagement in the Delta window, and the association was similar before the wave and smaller and imprecise afterwards, with no significant change between windows. This pattern is consistent with the MRT expectation that additional informational cues are valued under uncertainty, but it does not support a phase-dependent reversal of the richness effect. Because the indicator does not distinguish external links from native photos and videos, the result speaks to adding information to a tweet in general rather than to a specific medium. This limitation may also help explain why our result differs from Chen et al. (2020), who coded pictures and video specifically and studied a single official account.

5.4 Dialogic cues are not dialogue

H2 was not supported. Questions and hashtags were modestly negatively associated with engagement, and @mentions were unrelated to it. Studies of official government accounts found positive dialogic-loop effects (Chen et al., 2020; Zhang et al., 2022). The difference is plausibly one of source. When a health authority asks a question, it addresses an audience that expects it to answer; when an individual user does, the question may signal uncertainty or rhetorical challenge rather than an invitation to talk. Heavy hashtag use is also characteristic of promotional and low-effort posting. More fundamentally, the results fit the critique that counting questions, hashtags and @mentions does not measure dialogue (Taylor & Kent, 2014; Sommerfeldt & Yang, 2018; Wirtz & Zimbres, 2018). The features are cheap to add and are used for many purposes besides inviting conversation. Our evidence argues against treating them as levers for

engagement in user-generated crisis discourse; it does not test whether organizations that practise dialogue build stronger relationships.

5.5 What changes across crisis phases

Across the three windows, the associations for links and sentiment were stable in direction and did not differ significantly, whereas the dialogic cues shifted. The question penalty was largest before the Delta wave, and the @mention association moved from negative to positive over the three windows. This partly echoes the finding of Zhang et al. (2022) that engagement drivers differ across outbreak stages, but in our data the phase-sensitive elements are the interactional cues rather than informational richness. Because the windows also differ in corpus composition and contain collection gaps, we treat these shifts as descriptive, not as evidence of a causal effect of crisis phase.

5.6 Practical implications

Three implications follow for health communicators. First, linking to sources or attaching media and using a constructive, positive register are associated with modestly higher engagement, and these associations were positive in every window, although the link association was imprecise after the wave. Second, adding questions, extra hashtags or @mentions is not a reliable way to raise engagement in general vaccine discourse. Third, because reach dominates, investment in building and maintaining an audience, and in working with accounts that already have one, is likely to matter more than fine-tuning message features. For analysts who advise communicators, the main implication is to remove automated content and adjust for account reach before drawing any lessons from engagement data.

5.7 Limitations and future research

Several limitations qualify these findings. The data are observational, so the associations are not causal effects of message design. The corpus is a convenience sample assembled with undocumented query parameters, contains collection gaps and covers English-language tweets only. Engagement counts and account metrics reflect the undocumented time of collection rather than a fixed interval after posting, and replies and quotes are unavailable. Our automation rule detects only accounts identifiable from their posting application, so some automated content may remain. Sentiment was measured with a single lexicon-based tool, and the link indicator cannot separate external links from native media. The window boundaries follow the timestamps of the original analysis, so 30 June and 30 September fall outside the windows. Future work should combine account-level automation detection with message features, separate links from native photos and videos, compare official and non-official sources within the same model, and test dialogic theory with relational outcomes such as replies and sustained interaction rather than feature counts.

6. Conclusion

Large public tweet corpora are an attractive resource for learning what makes crisis messages resonate, but they need to be audited before they are modelled. In a widely available COVID-19 vaccine corpus, automated accounts produced a third of Delta-wave tweets and, together with omitted account reach, inflated the apparent effect of links or attached media more than threefold. After correction, informational cues and positive sentiment showed modest positive associations with engagement that were stable in direction, dialogic cues showed none of the positive effects that feature-based dialogic theory predicts, and account reach was the dominant predictor. Who speaks matters more than how a message is designed. Studies that ignore automation and reach risk attributing to message design what belongs to the messenger.

Declarations

Data Availability

The tweet corpus is publicly available on Kaggle under a CC0 licence (https://www.kaggle.com/datasets/gpreda/all-covid19-vaccines-tweets). All code needed to reproduce the analyses, including the automation rule, the model specifications and the figures, together with all model outputs, is available at https://github.com/AbdAllAh950/covid-crisis-communication-analysis.

Funding

This research received no specific grant. The first author's M.Sc. studies at ITMO University were supported by the Open Doors Russian Scholarship Project (2024).

Declaration of Conflicting Interests

The authors declare no conflicts of interest.

Author Contributions (CRediT)

Abdallah Sayed Ali Essa: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. Sergey V. Kovalchuk: Supervision, Methodology, Writing – review & editing.

Ethical Considerations

The study analyses a public, CC0-licensed corpus of public tweets. Results are reported only in aggregate; no individual users are identified, and automated accounts are described by their posting application rather than by account name. Under the guidelines governing secondary analysis of public, licensed data, ethical review was not required.

Declaration of Generative AI Use

During the preparation of this work, the authors used Claude (Anthropic) to assist with code review, re-analysis scripts and language editing. The authors reviewed and verified all analyses and text and take full responsibility for the content of the publication.

Acknowledgments

We thank the Faculty of Artificial Intelligence Technologies at ITMO University for the research environment and Assoc. Prof. O. V. Mitrenina for her review of the master's thesis on which this work builds.

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